

execute_cuda_attention()

DKV_CUDA_ANCHOR_ONLY = 1 ?

yes

anchor-only stub
(A/B regression)

no

stream_h2d (background)

issue the H2D restore for
the blocks of step $t+1$

cudaEventRecord(event_h2d)

fetch the routed blocks from HBM
cudaStreamWaitEvent(main, event_h2d)

*≈137 KB per block, ≈15 μs
on PCIe 4.0 — hidden behind
the previous decode step*

dkv_full_decode_kernel · dkv_decode.cu · -DGGML_CUDA=ON

anchor score + per-token delta projection (identical math to the Triton path)

K-side in-place residual scatter ← **res_K_positions**

V-side in-place residual scatter ← **res_V_positions**

split-K accumulation · scratch sized $H_q \times S_{split} \times d$, reallocated on model change

*DKV_CUDA_CHECK wraps every cudaMalloc, cudaMemcpy and kernel launch;
cudaGetLastError() is polled after each launch*

output tensor — same interface as the Triton path